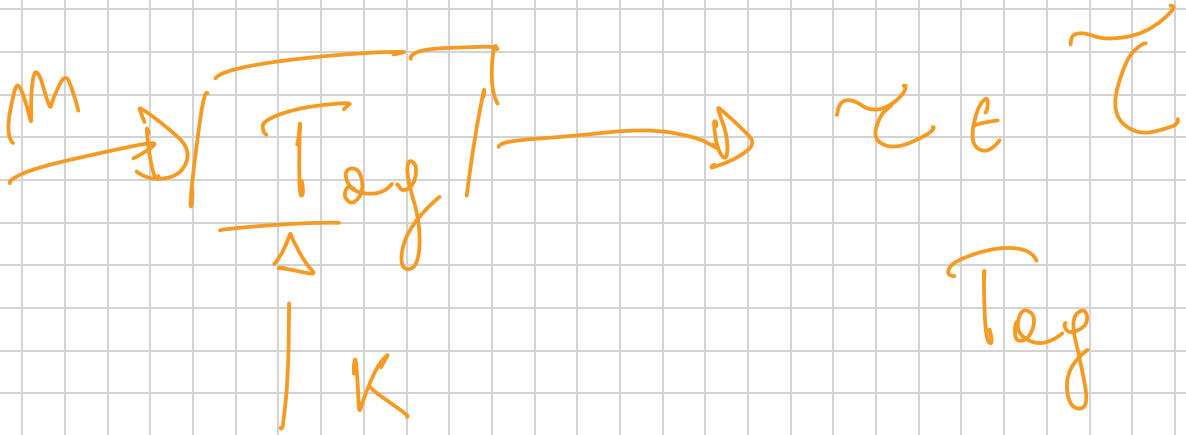
MESSAGE AUTHENTICATION CODES

We have already studied MACs in the statistical setting.
Today: The computational setting.



Tag is DETERMINISTIC.

DEF (UF-CMA) We say $\Pi = (\mathrm{Tag} : K \times M \to T)$ is UFCMA if $\forall$ PPT A

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$$\Pr\left[\mathrm{GAME}^{\mathrm{ufcma}}_{\Pi,A}(\lambda) = 1 \right] \leq \mathrm{negl}(\lambda).$$

$\mathrm{GAME}^{\mathrm{ufcma}}_{\Pi,A}(\lambda)$

A $\xrightarrow{m}$ $\tau = \mathrm{Tag}(k, m)$ $\xrightarrow{(m^*, \tau^*)}$ $\mathcal{C}$

$k \leftarrow K$

Output 1:

(1) $\tau^* = \mathrm{Tag}(k, m^*)$

(2) $m^* \notin \{m\}$

Remark:

- No protection against REPLAY ATTACKS. The application needs to remember this.

- Weak vs Strong UFCMA: We can allow $m^* \in \{m\}$ but then τ^* is FRESH.

Example. Consider $\text{Tag}'(k, m) = \text{all Tag}(k, m)$ s.t. Tag is deterministic. But Bob checks Tag and discards the first bit.

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In real schemes: Tags are unique. Then weak $\Rightarrow$ strong.

Let's construct efficient MACs.

TthM. Let $F = \{F_k : \{0,1\}^n \to \{0,1\}^n\}$ be a PRF family. Then, $Tag(k, m) = F_k(m)$ is UF-CMA.

Proof. Let $G(\lambda) \equiv G^{\text{MAC}}_{\Pi, A}(\lambda)$ and $H(\lambda)$ be the same as $G(\lambda)$ except that $F_k(\cdot)$ is now $R(\cdot)$ where

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$R \leftarrow R(n,n)$. By security of F:

$$G(\lambda) \stackrel{c}{\approx} H(\lambda).$$

All is left: $\forall A \quad \Pr[H(\lambda) = 1] \leq$ negl(λ). Even an A querying all of $\{0,1\}^n \setminus \{m^*\}$ can guess $R(m^*)$ w.p. $\leq 2^{-n}$ ☑

EXERCISE Then $\exists$ VTeg that is not a PRF. Insf say $\text{Tag}(k,m) = \text{OUT}_k(m)$

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Question: How To Tag a long message seq of length $N = m \cdot d$, $d \in \mathbb{N}$?

A few attempts: $m = m_1, \dots, m_d$

-) $\tau = \mathrm{Tag}(K, \bigoplus_{i=1}^{d} m_i)$

Not secure, e.g. given τ on $m = 0^n, 1^n$ output $\tau^* = \tau$ and $m^* = 1^n, 0^n$

-) $\tau = (\tau_1, \dots, \tau_d)$ and

$\tau_i = \mathrm{Tag}(K, m_i) \quad \forall i \in [d]$

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Permutations again !

→ Let's talk permutations :

$$\tau_i = \text{Teq}(K_1\ i \,\|\, M_i)$$

Look: given τ_1, τ_2 for M_1, M_2

Talk $m^* = (M_2, M_1)$ and $\tau^* = (\tau_2, \tau_1)$

$$\tau_2 = \text{Teq}(K_1, 2 \,\|\, M_2)$$

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Stell: Seien $c^{(1)} = (c_1, c_2)$ für $m^{(1)} = (m_1, m_2)$ und $c^{(2)} = (c_3, c_4)$ für $m^{(2)} = (m_3, m_4)$

Folge $m^* = (m_1, m_4)$

$c^* = (c_1, c_4)$.

Here is a deeper approach. Let

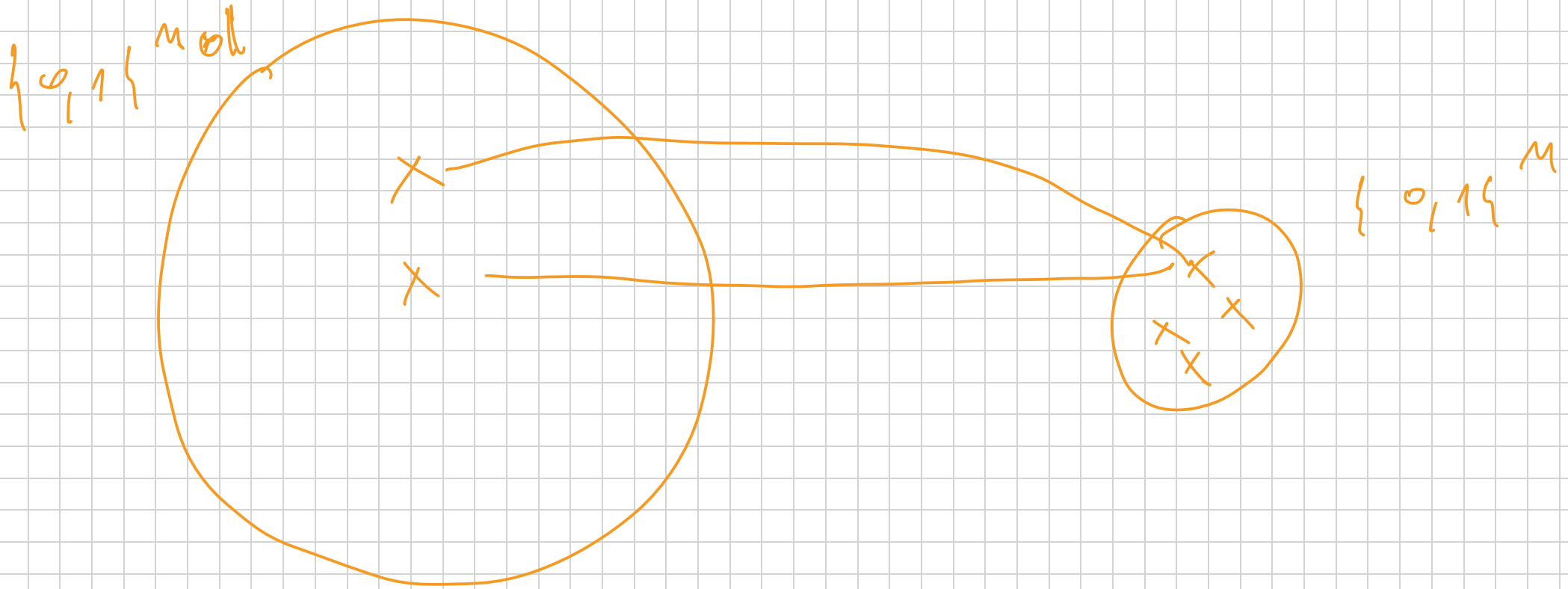
$$\mathcal{H} = \{ h_s : \{0,1\}^N \to \{0,1\}^M \}$$

given this, we can consider

$$F(\mathcal{H}) \equiv \{ F_k(h_s(\cdot)) \}_{k,s}$$

Q: what security from $\mathcal{H}$??

Notice that h_s is a shrinking function



Collisions ($m \neq m'$ s.t. $h_s(m) = h_s(m')$)

exist, but should be hard to find !

DEF (ALMOST UNIVERSAL). $\mathcal{H}$ is ε-AU

if $\forall$ $x, x' \in \{0,1\}^N$ s.t. $x \neq x'$

$$\Pr_{s}\left[h_s(x) = h_s(x') \right] \leq \varepsilon .$$

PERFECT UNIVERSAL : $\varepsilon = 2^{-n}$

We could also use "public-key" hash functions (e.g. SHA-3).

Remark: But those do not follow from OWF !

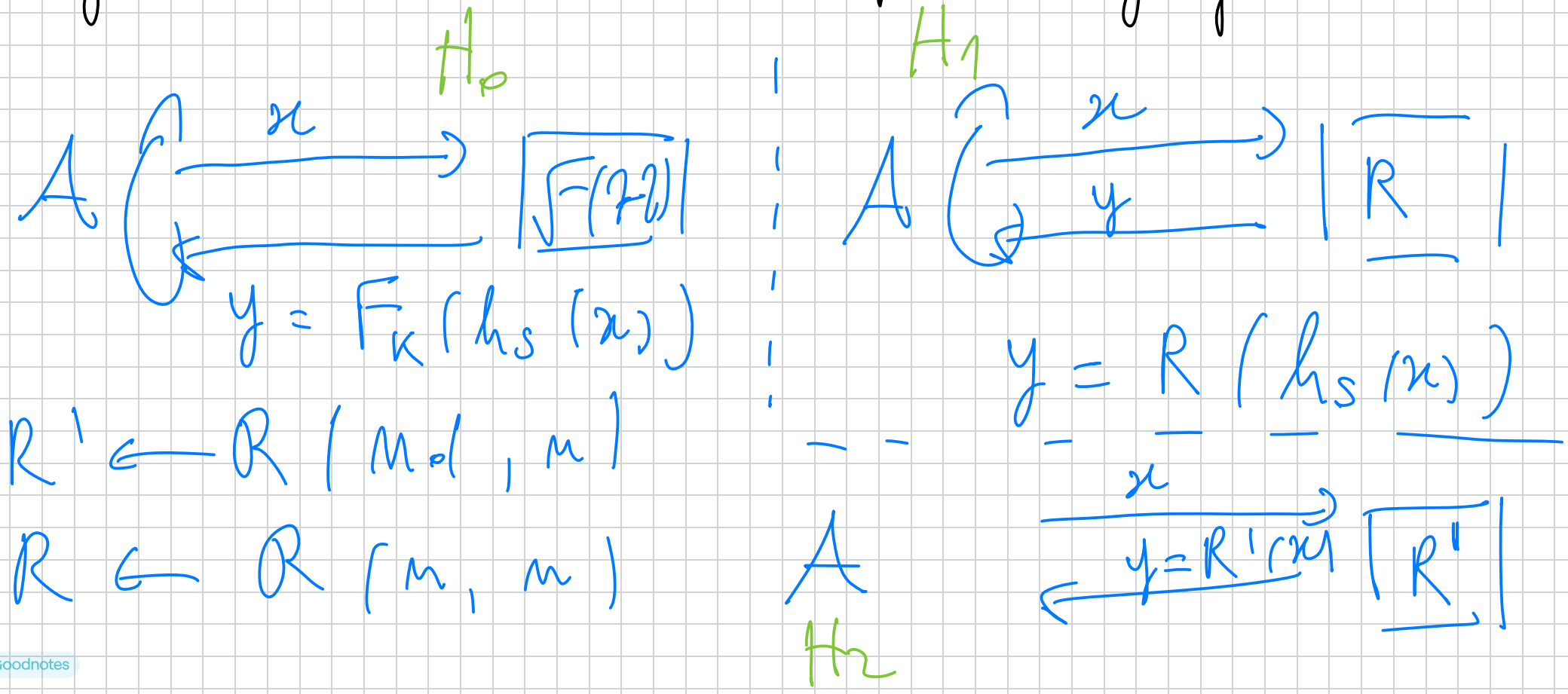
THM Assuming F is a PRF and H is AU, then $F(H)$ is a PRF.

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Cor. $F(\cdot,\cdot)$ is VFCMA or a MAC.

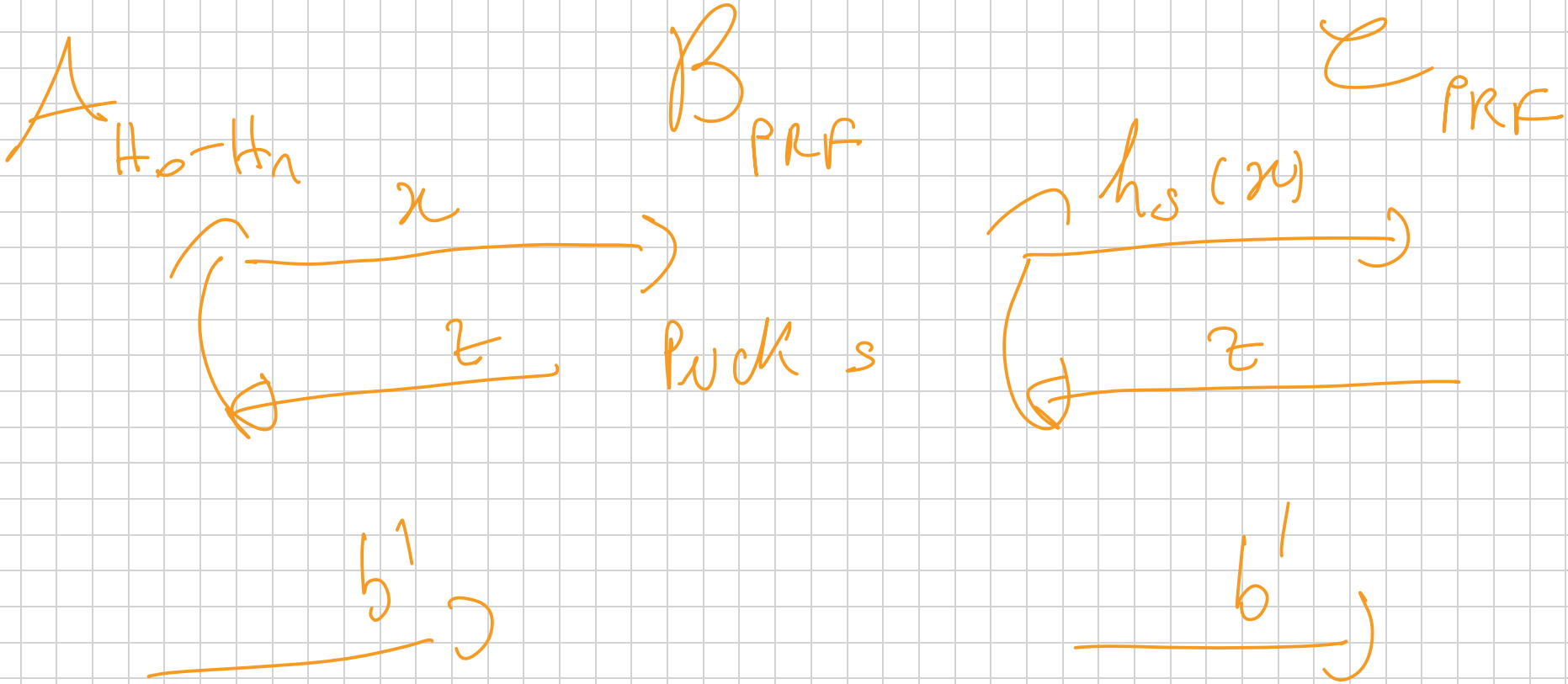
An important remark: For now, we only consider FIXED-INPUT LENGTH (FIL).

Proof. We consider a sequence of games:



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$H_0 \overset{\sim}{\approx}_c H_1$ (by PRF security)



The more frequently part is $H_1 \overset{\sim}{\approx}_s H_2$.

So long as # queries $q = \text{poly}(\lambda)$

Intuition: So long as the function R is queried on different points $h_s(x)$ then R produces a sequence of random and independent strings as R^1 does.

This suggests: BAD the event $\exists x_i, x_j$ s.t. $h_s(x_i) = h_s(x_j)$. So that:

$$SD(H_1, H_2) \leq Pr[BAD]$$

$$\leq \sum_{\substack{x_i, x_j \\ x_i \neq x_j}} Pr[h_s(x_i) = h_s(x_j)]$$

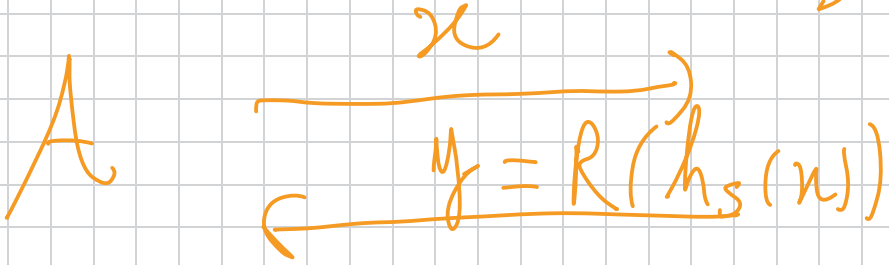
$$\leq q^2 \cdot \varepsilon \qquad \varepsilon \leq 2^{-n}$$

∎

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One needs to be a little careful how to define BAD:

$s \leftarrow \{0,1\}^\lambda$

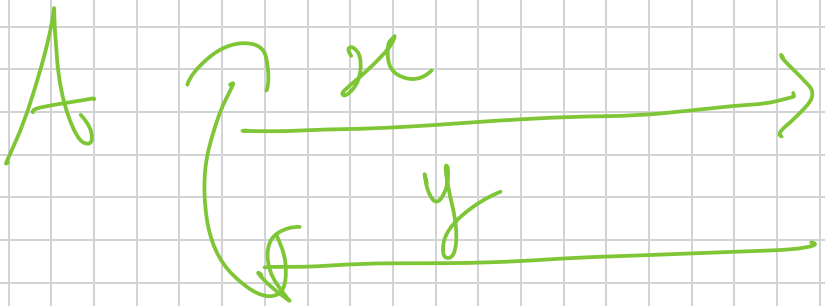


BAD

$\exists\, x_i, x_j$ s.t.

$h_s(x_i) = h_s(x_j)$

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$y \in \{0,1\}^m$

BAD:

$S \leftarrow \{0,1\}^\lambda$

$\exists\ x_0, x_1$ s.t.

$h_S(x_0) = h_S(x_1)$

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Let's construct h. Two examples:

*) $\mathbb{F} = GF(2^m)$, $m = m_1, \dots, m_d$

$m_i \in \mathbb{F}$ and $s = (a_1, \dots, a_d) \in \mathbb{F}^d$

$$h_{a_1 \dots a_d}(m_1, \dots, m_d) := \sum_{i=1}^{d} a_i \cdot m_i$$

($GF(2^m)$: given $m_i \in \{0,1\}^m$ make a polynomial e.g. $m_1 = 0101$

$0 \cdot X^3 + 1 \cdot X^2 + 0 \cdot X^1 + 1 \cdot X^0$)

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Fix $m = (m_1, \dots, m_d)$

$m' = (m_1', \dots, m_d')$

$\delta_i = m_i' - m_i$

Because $m \neq m'$, $\exists\ \delta_j \neq 0$. Wlog let $j = 1$. Collision iff:

$$\sum_i a_i m_i = \sum_i a_i m_i' \iff$$

$$a_1 \cdot \delta_1 = -\sum_{i=2}^{d} a_i \delta_i \iff$$

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$$a_1 = \delta_1^{-1} \cdot \left(- \sum_{i \neq 1} a_i' \delta_i' \right)$$

$$\Rightarrow \Pr\left[h_S(m) = h_S(m') \right] = 2^{-n}$$

PERFECT UNIVERSAL !

*) let's make Key shorter. $\mathbb{F} = GF(2^n)$

$S \in \mathbb{F}$.

$$h_S(m) = \sum_{i=1}^{\ell} m_i \cdot S^{i-1} = g_m(S)$$

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$$h_s(m) = h_s(m') \iff q_m(s) = q_{m'}(s)$$

$$\iff q_{m-m'}(s) = 0$$

$$\iff \sum_i (m_i - m'_i) s^{i-1} = 0$$

$m \cdot \deg = N.$

$$\Pr[h_s(m) = h_s(m')] =$$

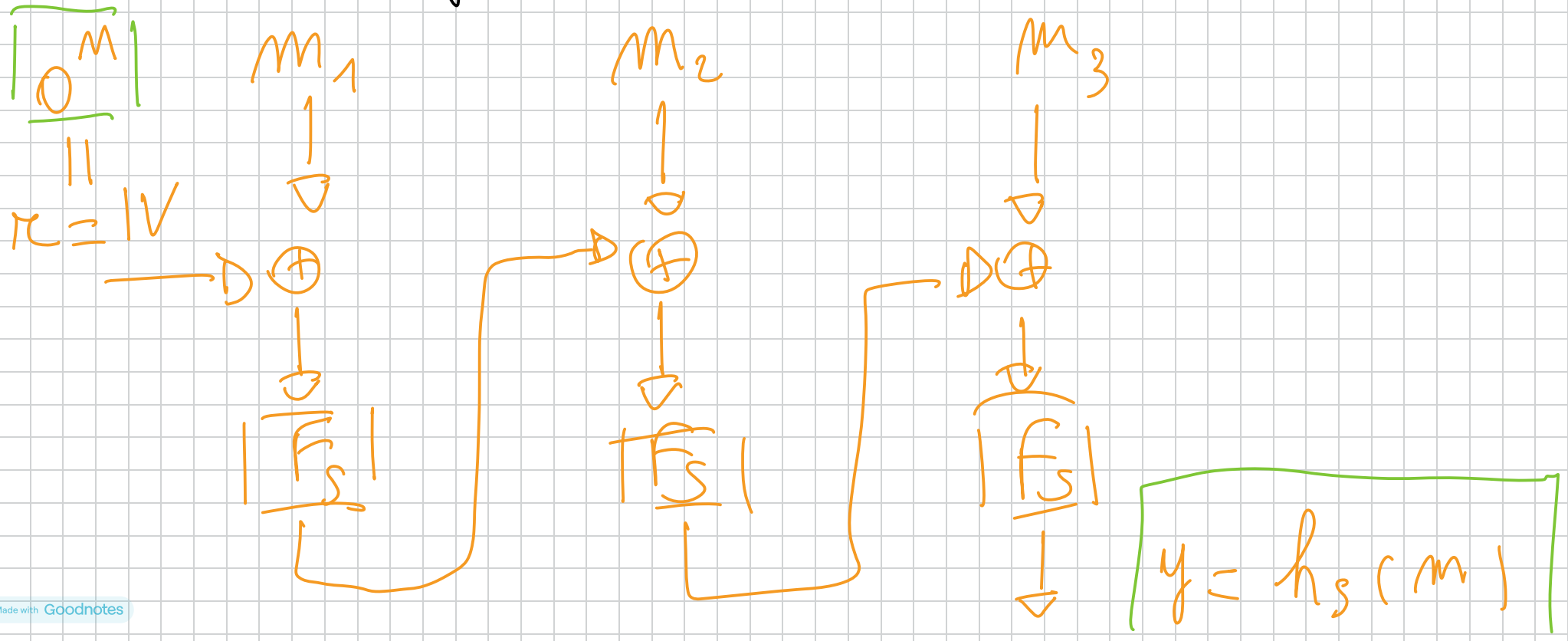
$$\Pr[s \text{ is a root of } q_{m-m'}]$$

$$\leq \frac{\deg - 1}{2^n} \approx \frac{N}{m \cdot 2^n}$$

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In the real world:

- CBC-MAC. This is just one $F(7\ell)$ s.t. both $\sqrt{F}$ and 7ℓ are built from a PRF F



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FACT "CBC MAC" as above gives a COMPUTATIONAL AU PRF.

Actually CBC-MAC is by itself as shown above a FIL UFCMA MAC.

EXERCISE. But not a VIL UFCMA MAC.

(Instead CBC-MAC as $F(PRF)$ is UFCMA for VIL.)

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We build a FORCER that asks queues of 1 block and forges on 2 blocks.

